

Building AI Agents in Enterprise

Week 1 - Lecture 2

What is Prompt Engineering?

- Prompt Engineering is the practice of **designing and crafting input instructions** to effectively communicate with AI language models, optimizing the prompts to achieve desired **outputs with greater accuracy, relevance, and consistency**.
- **Bridge Role:** Acts as the interface between human intent and machine reasoning.
- Fundamental to all LLM applications
- **Evolution:** From simple keyword search → instruction prompts → few-shot reasoning → automated optimization.

Core Principles of Prompt Engineering

- **Clarity and Specificity:** Write clear, detailed instructions that specify exactly what you want the AI to do, including format, tone, length, and any constraints or requirements for the response.
- **Context and Examples:** Provide relevant background information and use concrete examples to illustrate your expectations, helping the AI understand the task better and generate more accurate outputs.
- **Iterative Refinement:** Treat prompt engineering as an experimental process where you test, analyze results, and continuously refine your prompts based on what works best for your specific use case.
- **Strategic Structure:** Organize prompts logically with techniques like step-by-step instructions, role assignments, or structured formats to guide the AI through complex reasoning or multi-part tasks.

Examples

- **Bad Example - Vague and Unclear:** "Write something about cats."
- **Good Example - Clear and Specific:** "Write a 200-word informative paragraph about indoor cat care for new pet owners. Include specific tips on feeding schedules, litter box maintenance, and creating a safe environment. Use a friendly, helpful tone."
- **Bad Example - Ambiguous Instructions:** "Make product descriptions better."
- **Good Example - Detailed Requirements:** "Rewrite this product description to be more compelling for online shoppers. Make it 150-200 words, highlight the main benefits in the first sentence, include specific technical specifications, and end with a clear call-to-action. Target audience is tech-savvy professionals aged 25-40."
- **Bad Example - Multiple Unclear Requests:** "Help me with my presentation and make it good and professional."
- **Good Example - Specific, Actionable Tasks:** "Review my 10-slide sales presentation and provide feedback on: 1) slide organization and flow, 2) visual design consistency, 3) clarity of key messages, and 4) strength of the closing call-to-action. Focus on improvements for a 30-minute executive audience presentation."

Instruction Structuring

- **Definition:** Breaking down a prompt into *role*, *task*, *context*, and *output format*.
- **Why:** Helps the LLM understand *who it is*, *what to do*, *with what info*, and *how to present it*.
- **Core Elements:**
 - **Role:** Define perspective (e.g., “Act as a financial analyst”).
 - **Task:** State action clearly (e.g., “Summarize Q2 earnings”).
 - **Context:** Add background (e.g., “Company is a retail chain expanding into e-commerce”).
 - **Output Format:** Specify delivery (e.g., “In 3 bullet points, with one risk highlighted”).
- **Benefit:** Reduces ambiguity → increases consistency.

Example:

Wrong: “*Summarize the report.*”

Correct: “*You are a financial analyst. Summarize the attached Q2 retail earnings report in 3 bullet points, focusing on revenue growth, customer acquisition, and risks.*”

Controlling Outputs

- **Definition:** Guide the LLM to structure responses in predictable, machine-readable, or human-friendly ways.
- **Why:** Prevents overly verbose, inconsistent, or unstructured answers.
- **Techniques:**
 - Ask for **specific formats**: tables, bullet points, numbered lists.
 - Require **machine-readable formats**: JSON, XML, CSV.
 - Set **length/verbosity constraints**: “Answer in 3 sentences” or “limit to 50 words.”
- **Common Use Cases:** Data extraction, summaries for dashboards, structured outputs for downstream code.
- **Benefit:** Makes outputs reliable, reusable, and easier to integrate into applications.
- DSPy is a very good package to control the output structure

Example (Prompt only):

Wrong: *“Tell me about Paris.”*

Correct: *“Give me 3 bullet points about Paris: history, culture, and food.”*

Zeroshot and FewShot Prompting

Zero-Shot Prompting

- *Definition:* Ask the model to perform a task **without examples**.
- *When useful:* Quick tasks, general knowledge, flexible queries.
- *Example:*
 - Prompt: “*Classify the sentiment of this review: ‘The delivery was late and frustrating.’*”
 - Output: “*negative*”

Few-Shot Prompting

- *Definition:* Provide **a few labeled examples** in the prompt to guide the model.
- *When useful:* Ambiguous tasks, domain-specific tone, structured outputs.

Chain of Thought Prompting

Definition: A prompting technique where the model is asked to **reason step by step** before giving the final answer.

Why use it:

- Improves accuracy on complex reasoning tasks (math, planning, multi-step analysis).
- Reduces errors caused by skipping intermediate reasoning.
- Makes the model's thought process **transparent and interpretable**.

How to implement:

- Add phrases like: *“Let's think step by step”* or *“Explain your reasoning before answering.”*
- Can combine with few-shot prompting for higher reliability.

Negative Prompting

Definition: Technique of explicitly telling the LLM **what *not* to do** in its response.

Why use it:

- Reduce unwanted behaviors (hallucinations, verbosity, irrelevant info).
- Improve control over tone, style, and accuracy.
- Helpful in safety, compliance, and brand voice alignment.

How to implement:

- Add constraints like *“Do not include personal opinions”* or *“Avoid jargon”*.
- Use with positive instructions for balance.

System Prompt vs User Prompt

System Prompt (Role / Instruction Prompt)

- *Definition:* Sets the **behavior, role, or style** of the model for the session.
- *Purpose:* Guides overall tone, persona, or rules the LLM should follow.
- *Example:*
 - "You are an expert retail analyst. Always provide concise, data-driven answers in bullet points."
- Often set once at the **start of a session**.

User Prompt (Task / Query Prompt)

- *Definition:* Specifies the **actual task or question** the user wants the model to perform.
- *Purpose:* Drives the specific output for the current request.
- *Example:*
 - "Summarize the sales performance of Product X for Q2 in 3 bullet points."
- Can be sent **multiple times**, usually interacting with the system prompt context.

Prompt Chaining

- **Definition:** Breaking a complex task into smaller steps, where the **output of one prompt becomes the input to the next.**
- **Why use it?**
 1. Handles complex reasoning better
 2. Reduces hallucinations
 3. Makes outputs more interpretable and reusable

Example (E-commerce):

- Prompt 1: *“Summarize this customer review in one sentence.”*
- Prompt 2: *“Classify the sentiment of this summary as positive, negative, or neutral.”*
- Prompt 3: *“Suggest one product improvement based on the review.”*

Chaining makes LLMs act like a **pipeline of reasoning**, not a single-shot answer.

Key Takeaways

- **Prompt Engineering = Communication Skill**

- Craft clear, structured, and context-rich instructions to get accurate outputs.

- **Core Patterns**

- Zero-shot & Few-shot for flexibility vs. accuracy
- Chain-of-Thought for reasoning
- Negative prompting for guardrails
- System prompt (role) + User prompt (task) = Better control

Key Takeaways

- **Advanced Techniques**

- Prompt Chaining → break down complex tasks
- Automatic Prompt Optimization → scale beyond manual tuning
- Prompt Management → versioning, governance, and reproducibility

- **Mindset Shift**

- Treat prompts as **production artifacts**, not one-off instructions.

Let us build a Inventory Planning Agent

Problem:

A mid size wholesaler has 500+ SKUs across 3 warehouses.

- 500+ SKUs across 3 warehouses
- Hours spent manually checking stock levels
- Applying reorder rules by hand
- Stock-outs = lost sales
- Overstocking = tied-up capital

Constraints:

- Keep the system costs super low.
- I do not want to host the system on prem because I want to scale based on demand.

